PACMAN Legacy Firmware (1.X)

Branches, build steps and bitwise reproduction

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1 Purpose and scope

PACMAN firmware comes in three orthogonal families: *legacy* (1.X, everything before consistent versioning), *hijinks* (2.X, the 2024 firmware for the 1v5 test stands, described in a companion note) and *zipline* (3.X, the current firmware in the `pacman-firmware` and `pacman-software` repositories). This note describes how the legacy firmware is kept in the repository `mulhearn/pacman`: one branch per hardware line, each holding as much of that line's firmware as possible, every state that was released or deployed reproducible from the repository alone, and the proof of each reproduction.

A bitstream is *reproduced bitwise* when the SHA-256 of its configuration body (the bitstream without its header, which carries the build date and paths) equals that of the reference bitstream. The bodies are quoted here by their first eight hex digits.

2 Layout

- One branch per hardware line (`hw1v3`, `hw1v4`, `hw1v5`), firmware only, as linear as the history allows.
- Historic commits keep their original hashes. New commits go only at a branch's tip:
  - *snapshots*: firmware that was built outside the repository, restored from its Vivado project archive;
  - *build steps*: what makes a state build from the repository alone (fixes without behavior change, staging scripts); they stay on the branch;
  - *build options*: variants of a build (rot8).
- Superseded side lines, which change nothing on the main lines, are recorded on two *excursions* branches, firmware and software.
- Branches under review are named `working-on-`*name* until they are frozen (renamed without the prefix and tagged); they may be rewritten only until then.
- A checker (`check-legacy-hw1vN.sh`, run on fresh clones) enforces the layout: each spine ends at its historic tip with the original hashes and is followed by exactly the listed new commits, the build steps change only build scripts, the excursions never change the tree, and commit messages cite only commits on the spines.

The first commit of all of them is 0b9bee4 (2019-11-16, the Trenz TE0720 skeleton). Zipline shares only that commit with legacy; `pacman-firmware` and `pacman-software` share no history with this repository.

Branch	Tip	Historic tip	New at the tip
`working-on-legacy-firmware-hw1v3`	9fbc1dc	501180f	893 snapshot, build step, 883 snapshot
`working-on-legacy-firmware-hw1v4`	d08d199	21019ee	894 (two commits), build step
`working-on-legacy-firmware-hw1v5`	7062e17	2425e67	SYNC FIX build step, build option rot8
`working-on-legacy-firmware-excursions`	d5fd1c7	(2425e67)	three record merges
`working-on-legacy-software-excursions`	4a27ae1	(2425e67)	three record merges

Table 1: The legacy branches (2026-10-31).

3 The hw1v3 line

Spine. The trunk (v1.2, v1.2b, three 2020 merges), then the v1.3 releases: v1.3-0.0 (a10a44a, 2020-12-23, the first v1.3 board), v1.3-0.1 (e26918e, external sync and trigger inputs), v1.3-0.2 (1dfc351, no firmware change), v1.3-0.3 (2099d0d, POSI/PISO invertible), and v1.3-1.0 (501180f, 2022-05-16, Peter Madigan's adjustable MCLK). Every tag builds as committed with Vivado 2018.3 (the first run fails at IP generation, the rerun builds).

New commits.

- **893** (a5f8e91): Armin Karcher's V3 firmware (2024-02), the version that runs on the 1v3 boards (version register `0x37D`), from its project archive `pacman-fw893.xpr.zip`: the archive's project Tcl as `tcl/pacman-fw15.tcl` (Vivado 2021.1) and the sources it needs that differ from 501180f (`clk_ddr_en.vhd`, `uart_en.vhd` new; `larpix_mclk_sel.vhd`, `larpix_reset_gen.vhd` changed; the FIFO `.xci` as upgraded to 2021.1).
- **Build step** (1f66dcd): `tcl/build_fw15.sh` and `tcl/build_fw15.tcl`.
- **883** (9fbc1dc): 893 with edge-sensitive SYNC/TRIG inputs (Armin, 2024-05, for the 2x2 trigger module): `larpix_reset_gen.vhd` and that archive's project Tcl.

Build. From the top of a checkout, with the Vivado 2021.1 settings sourced:

```
bash tcl/build_fw15.sh        # -> products/zsys_wrapper.bit
```

The project Tcl is Armin's export, unchanged: it names its files in the layout of his Vivado project and has no top-level wrapper. The script stages the files it requires (its own `checkRequiredFiles` list) from `srcs/`, `constraints/` and `srcs/ip` into that layout under `build/fw15`, runs the export, adds the wrapper as `tcl/build_xpr.tcl` does, and builds through `write_bitstream`.

4 The hw1v4 line

Spine. From v1.3-0.3: the rev4 commits (new constraints with one GLOBAL_SYNC, I2C virtual registers, eMMC), v1.4-0.0 (4532a23, 2022-04-12), the merge of v1.3-1.0 (b37e07f), v1.4-0.2 (0681491, 2022-10-04, the last larpix release), and Armin's module2 upload (af8c85b, 21019ee, 2023-03-01: the resethack 1.8 project Tcl `tcl/pacman_resethack1.8.tcl` and `pacman_r4.xdc`).

New commits.

- **894**, resethack 1.8 (a693c5e, c98f6b0): the sources the module2 Tcl needs that were never committed (`uart_en.vhd`, `tristate_out.vhd`, `clk_ddr_en.vhd`, `larpix_reset_gen.vhd` with the per-tile reset mask, `zsys_wrapper.vhd`), from the project archive `pacman_resethack1.85.xpr.zip`, and the FIFO `.xci` as upgraded to 2021.1.

- **Build step** (d08d199): `tcl/build_resethack1.8.sh`, `tcl/recreate_resethack1.8.tcl`, `tcl/build_resethack1.8.tcl`.

Build. With the Vivado 2021.1 settings sourced:

```
bash tcl/build_resethack1.8.sh     # -> products/zsys_wrapper.bit
```

The recreate script stages every file Armin's Tcl requires under `build/resethack1.8`, links the Windows path by which it imports the block design (relative on Linux) to the staged tree, and sources a copy with one edit: Vivado 2021.1 places `pacman_r4.xdc` in `constrs_1/imports/constraints/`, so the target constraints path is changed to match. The build launches the implementation a second time when the first launch stops at the 2021.1 IP generation.

5 The hw1v5 line

Spine. From v1.4-0.0: the 2024 work (c2872b5, new constraints, IP upgrade, Vivado 2020.2; 36e32af, SYNC FIX: RESETN replaced by a per-tile SYNC; 5588b58, constraints cleanup; d8790e0, rot8 constraints as a separate file), ending at 2425e67 (2024-08-11). 2425e67 does not build as committed.

New commits.

- **Build step** (3155d4a, "restore build from repo"): four fixes without behavior change: `build_xpr` writes the hardware description with `write_hwdef` (Vivado 2020.2 writes no `.sysdef`); the data_tx subset converter's options are set after `validate_bd_design` (AR 58038); `larpix_mclk_sel` has integer generics, so the block design regenerates; `larpix_clk`'s `C_RW_REG3_DEFAULT` is `0x00000000`, the value always built (the source's `0X00000004` was parsed as 0). The hijinks firmware starts here.
- **Build option rot8** (7062e17): `pacman-rot8.xdc` copied onto `pacman.xdc` (tiles 1–8 on tiles 2–10), as the 2024 rot8 builds were made.

Build. With the Vivado 2020.2 settings sourced: `tcl/recreate_xpr.tcl`, then `tcl/build_xpr.tcl`.

6 Excursions

Each excursion is recorded by one `git merge -s ours` on 2425e67: the excursion's tip becomes the merge's second parent, so all its commits keep their original hashes, while the tree never changes (nothing is combined). The first-parent log is the index; a single-branch clone holds every excursion.

Branch	Excursion	Owner	Base	Tip	What
firmware	pacman-v1.2b-1MHz	Peter Madigan	v1.2b-0.0	ba2ef0a	1 MHz MCLK for the 1v2b board
firmware	pacman-v1.3-1MHz	Peter Madigan	v1.3-0.2	38af5a7	fixed 1 MHz MCLK
firmware	pacman-v1.3-5MHz	Peter Madigan	v1.3-0.2	8a7e990	fixed 5 MHz MCLK
software	pacman-v1.2b (v1.2b-0.1)	Peter Madigan	v1.2b-0.0	6d70ed2	maintenance release (packaging)
software	pacman-1v4-petalinux20.2	Armin Karcher	v1.4-0.2	5e67360	PetaLinux 2020.2 port, eMMC boot
software	legacy-pacman-1v4-pe	Michael Mulhearn	(the port)	469c948	+ I2C fix for the MUX: the best 1v4 server

Table 2: The excursions. The fixed-MCLK lines are superseded by v1.3-1.0's adjustable MCLK.

7 Bitstreams and their reproduction

Bitstream	Body	Commit	Vivado	Reproduction
v1.4-0.2 release	1362d6b3	0681491 (tag)	2018.3	bitwise, the tag as committed (2026-10-13); again from a fresh clone (2026-11-02)
893 (1v3 boards)	3194812d	a5f8e91 + build step	2021.1	bitwise from a fresh clone with the in-repo build step (2026-10-31); reference read from a running 1v3 board
883	0ff4bf66	9fbc1dc	2021.1	bitwise from a fresh clone with the in-repo build step (2026-10-31)
894 (resethack 1.8)	ec9ecc71	c98f6b0 + build step	2021.1	bitwise from a fresh clone with the in-repo build step (2026-10-31); reference: the "84"/r8hack boot images
SYNC FIX	4fdfb7d2	3155d4a	2020.2	bitwise from a fresh clone with the in-repo build step (2026-11-01); reference: LBNL 1v5_20240731A_ucd_syncfix
SYNC FIX rot8	5fdf3d45	7062e17	2020.2	bitwise from a fresh clone with the in-repo build option (2026-11-01); reference: LBNL 1v5_20240810A_ucd_rot8
"74" (2x2 image)	f0b74071	—	—	not reproduced: built outside the repository, no project archive

Every bitstream above except "74" was built once more from a fresh clone of the repository with its own scripts, bitwise (finished 2026-11-02). Pending: the tags without an artifact (v1.3-0.0 to 0.3, v1.3-1.0, v1.4-0.0) built once and their bodies recorded; v1.3-1.0 and v1.3-0.3 compared with 60209db9, the bitstream of a 2022-10-31 BOOT.BIN found deleted on the 2x2 run-1 SD card, which would give the 1v3 line its release reference.

8 Conventions

- Historic commits keep their author, dates and hash. The snapshots, build steps, build option and excursion merges are authored by augie; the hijinks 2.X restores by Michael Mulhearn.
- Work brought in from elsewhere enters as a new commit, described by content: no cherry-pick lines, no repository-qualified branch names. Bitstream hashes and bundle names are data and may appear.
- A snapshot changes only the files its archive changes; a build step changes only build scripts and fixes without behavior change; each records its bitwise result in its message.

9 Old branches

Deleted on 2026-10-31, after a check that every commit they held is reachable from a kept branch or tag, or has its tree there: `legacy-firmware-historic`, `working-on-legacy-firmware-buildable` (the earlier

design with build loops), the old excursions record, `hijinks-hw1v5-2020.2`, `legacy-pacman-1v4-pe`, `pacman-server-hijinks-2018.3`, and the tags `sync-fix` and `hijinks-2v8`. The only commit let go with a tree of its own is the first 2.8 restore (61e2fa4): the kept 2.8 plus the files its successor removes, which equal 2425e67's. Six further branches (the 2020.2 GPIO-only hwcheck, the 2024 v1.5 reorganization, the 2020.2 and 2023.2 hardware demos, a 3v4 driver fixup) are under discussion.

10 Still open

Freezing (the branches renamed without `working-on-`), and annotated tags on every state of interest, each carrying its build result (toolchain, how it builds, configuration body and build-product hashes), under a prefix of our own; a single `legacy` branch merging the line branches is under discussion.